Fatima Safwa

Computer Science (AI & ML) Student | Full-Stack & AI Developer
Phone: +91 8217818290 | Email: fatimasafwa30@gmail.com | github.com/fatimasafwa30 | linkedin.com/in/fatima-safwa-08097b33b

Professional Summary

Computer Science (AI & ML) student interested in software development and applied AI. Experience building full-stack applications, backend APIs, agentic AI systems, and geospatial interfaces using React, Next.js, Node.js, Python, and Supabase.

Education

Anjuman Institute of Technology and Management — Bangalore, India
Bachelor of Engineering (B.E.), Computer Science (AI & ML) — *October 2024 – July 2028*

Experience

Backend Developer – Noob Hackverse College Hackathon Platform — *2026 – Present*
Next.js, Supabase, PostgreSQL, REST APIs, Zod

- Developing the backend for a college hackathon platform using Next.js Route Handlers and Supabase.
- Designed PostgreSQL tables for user profiles, teams, team members, and event configuration.
- Built APIs for authentication, user sessions, team creation, team management, and member operations.
- Added Zod validation, role-based access, authentication checks, and rate limiting to secure API routes.
- Created API documentation and health-check endpoints to support testing and development.

Projects

Support Intelligence Agent – Agentic AI Customer Support System — *September 2026*
Python, Scikit-learn, TF-IDF, Logistic Regression, Retrieval

- Built an AI support agent that classifies customer requests, retrieves relevant resolutions, generates grounded responses, and decides when a case should be escalated.
- Trained a 15-intent classifier using TF-IDF and Logistic Regression with confidence-based abstention for uncertain requests.
- Added retrieval and confidence checks so uncertain queries can be passed to a human instead of receiving an unreliable response.
- Evaluated the classifier on a leakage-safe validation split, achieving 87.77% accuracy and 85.59% macro-F1 at a 0.25 abstention threshold.

SAMUDRA SAKSHI – Oil Spill Detection & Vessel Attribution Platform — *Smart India Hackathon 2026*
React, Vite, React Router, Leaflet, React-Leaflet, JavaScript

- Built the frontend for an oil-spill detection and vessel-attribution platform with 7 connected interfaces covering detection, characterization, drift analysis, attribution, evidence generation, and evaluation.
- Added interactive geospatial maps to visualize detected slicks, vessel tracks, drift paths, origin regions, and attribution results.
- Integrated abstention and evidence-state logic across the detection, attribution, and evidence interfaces so low-evidence cases are handled consistently instead of forcing an attribution.
- Improved evidence provenance and packet integrity by replacing hard-coded values with case-derived data and generating a deterministic SHA-256 integrity hash from the case payload.
- Added verification checks for evidence generation and removed unsupported confidence and provenance values from the interface.

Repo Intelligence – GitHub Repository Analysis Tool — *September 2026*
React, Vite, Express.js, Octokit, Groq API

- Built a full-stack tool that analyzes GitHub repositories and explains their structure, major components, and setup.
- Developed an Express.js backend and used Octokit to retrieve repository data from GitHub.
- Connected the backend to the Groq API through an /api/explain endpoint to generate repository explanations.
- Created the React/Vite interface where users can enter a GitHub repository and view the generated analysis.

Technical Skills

Languages: Python, JavaScript, HTML, CSS
Frontend: React.js, Next.js, Vite, React Router, Leaflet.js, React-Leaflet, OpenStreetMap, Responsive Design
Backend: Node.js, Express.js, Next.js Route Handlers, REST APIs, WebSockets, CORS
AI/ML: Scikit-learn, TF-IDF, Logistic Regression, Intent Classification, Retrieval, Agentic AI
Databases: PostgreSQL, MongoDB, Supabase
Security: JWT, Role-Based Access Control (RBAC), Zod, Rate Limiting
APIs & Tools: Groq API, Octokit, Git, GitHub, npm, API Testing
Cloud & Deployment: Vercel, Render, CI/CD